

Projective Space

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Localization of $C[x]$

Let $C[x]$ be the ring of polynomials with complex coefficients. We consider three types of localization:

- Localization at a prime ideal $(x - a)$, where $a \in C$.
- Localization at the zero ideal (0) .
- Localization at a specific polynomial $f \in C[x]$.

1. Localization at $(x - a)$

For a point $a \in C$, the prime ideal $(x - a)$ in $C[x]$ allows localization to the ring $C[x]_{(x-a)}$. Elements in this ring are of the form:

$$\frac{g(x)}{h(x)} \quad \text{where } g(x), h(x) \in C[x] \text{ and } h(a) \neq 0.$$

2. Localization at the Zero Ideal (0)

Localization at the zero ideal (0) corresponds to the field of fractions $C(x)$, consisting of all rational functions:

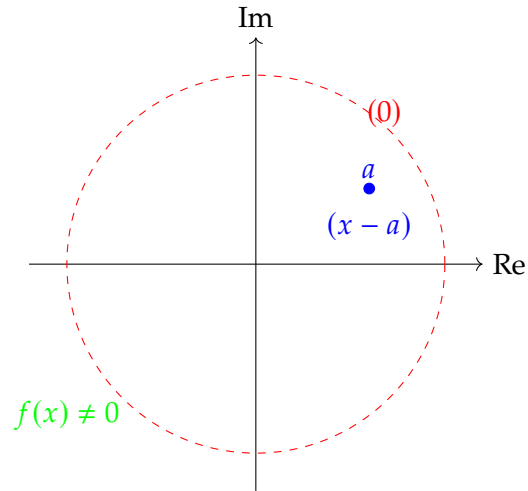
$$\frac{g(x)}{h(x)} \quad \text{where } g(x), h(x) \in C[x] \text{ and } h(x) \neq 0.$$

3. Localization at a Polynomial f

For $f \in C[x]$, the localization $C[x]_f$ consists of elements:

$$\frac{g(x)}{f^n} \quad \text{where } g(x) \in C[x] \text{ and } n \in \mathbb{N}.$$

Graphical Representation in $\mathbb{C}$ (Complex Plane)

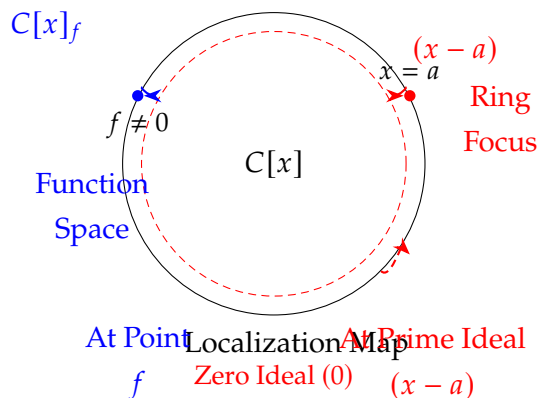


In the diagram:

- The blue point represents the localization at $(x - a)$, where we focus on values near $x = a$.
- The red dashed circle represents localization at the zero ideal (0) , which includes all points in $\mathbb{C}$.
- The green label represents areas where $f(x) \neq 0$ for localization at a polynomial f .

Localization in $C[x]$

Let $C[x]$ denote the commutative ring of complex polynomials. The localization at a point f , a prime ideal $(x - a)$, and the zero ideal (0) reflects function spaces within $C[x]$.



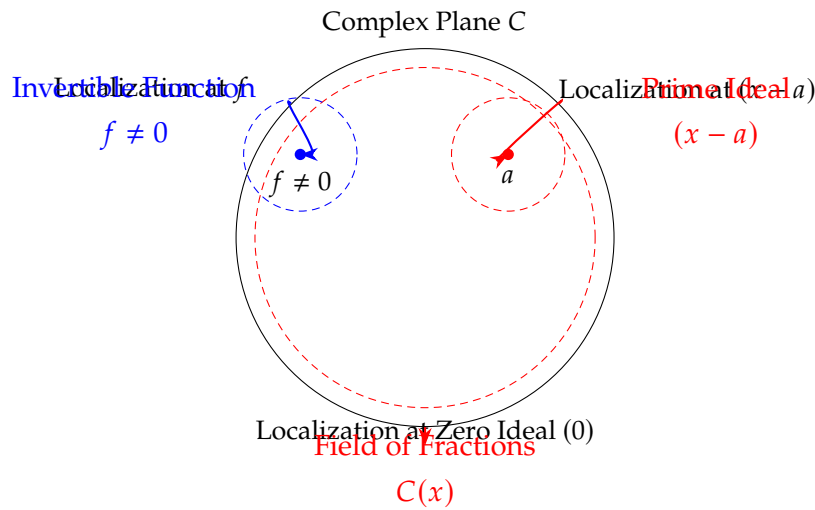
Explanation

In this diagram:

- The circle labeled $C[x]$ represents the function space of complex polynomials.
- The red dashed circle surrounding the origin corresponds to the zero ideal (0) , where we obtain the field of fractions $C(x)$.
- The red dot $x = a$ represents localization at the prime ideal $(x - a)$.
- The blue dot represents localization at a point f (where $f \neq 0$), showing a ring where f becomes invertible.

This setup represents the localization as focusing on specific points or subspaces within the function space $C[x]$, reflecting the commutative nature of polynomial functions over complex values.

Illustration of Localizations in $\mathbb{C}[x]$

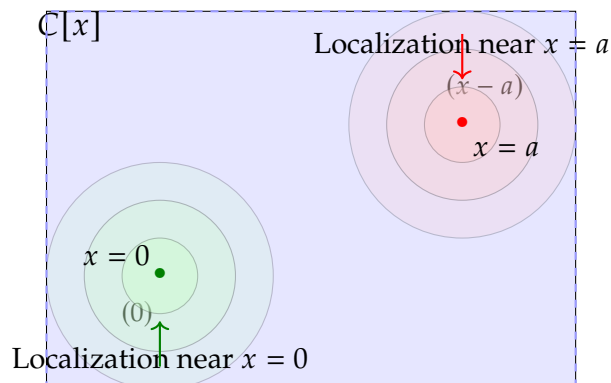


Explanation of the Illustration

In this illustration:

- The large circle labeled C represents the complex plane as a function space.
- **Red Dashed Circle (Zero Ideal Localization)**: Surrounding the origin, representing the field of fractions $C(x)$, which has rational functions accessible across the plane.
- **Red Circle at $x = a$** : Localization at the prime ideal $(x - a)$, focusing on behavior around $x = a$.
- **Blue Circle around f** : Localization at a function f , where $f \neq 0$, indicating regions where f can be inverted.

Each arrow represents a "focus" that restricts attention to specific aspects of the function space.



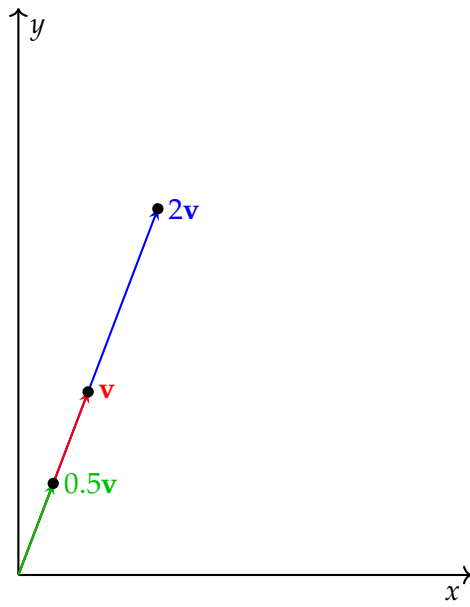
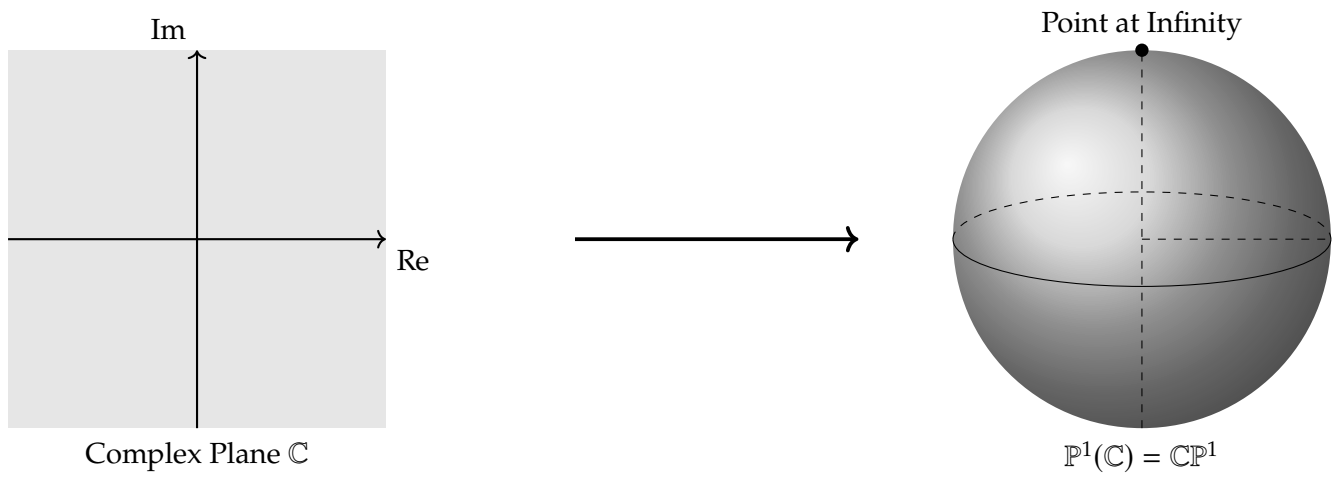


Figure 1: In $\mathbb{R}^2$ (or $\mathbb{C}^1$), vectors $\mathbf{v}$, $2\mathbf{v}$, and $0.5\mathbf{v}$ lie on the same line.



Projective Space

Definition. Let $\mathbb{K}$ be a field and $n \geq 0$. The **n-dimensional projective space** over $\mathbb{K}$, denoted $\mathbb{P}(\mathbb{K}^{n+1})$, is defined as the quotient:

$$\mathbb{P}(\mathbb{K}^{n+1}) := (\mathbb{K}^{n+1} \setminus \{\mathbf{0}\})/\sim,$$

where $\sim$ is an equivalence relation defined on $\mathbb{K}^{n+1} \setminus \{\mathbf{0}\}$ by

$$\mathbf{v} \sim \mathbf{w} \iff \exists \lambda \in \mathbb{K} \setminus \{0\} \text{ such that } \mathbf{v} = \lambda \mathbf{w}$$

for $\mathbf{v}, \mathbf{w} \in \mathbb{K}^{n+1}$.

Remark. $\mathbb{P}(\mathbb{K}^{n+1})$ is the set of lines through the origin in $\mathbb{K}^{n+1}$. These lines represent equivalence classes of non-zero vectors in $\mathbb{K}^{n+1}$, where two vectors are equivalent if they are scalar multiples of each other.

Note (Notation). The projective space $\mathbb{P}(\mathbb{K}^{n+1})$ has dimension n .

- $\mathbb{P}(\mathbb{K}^{n+1}) = \mathbb{P}^n(\mathbb{K})$
- $\mathbb{P}(\mathbb{K}^{n+1}) = \mathbb{K}\mathbb{P}^n$